



INFLUENCE OF COW DUNG AND UREA FERTILIZER ON SOIL pH, OC, TN AND MICROBIAL POPULATION UNDER MAIZE POT EXPERIMENT IN A SAVANNA ALFISOLS

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ARTICLE INFO.

ABSTRACT

Key words:

cow dung, urea fertilizer, microbial population, maize, bacteria, fungi, OC, TN and pH

This experiment was conducted in the Screen house at the Department of Soil Science, in Ahmadu Bello University Zaria, Kaduna state to evaluate the influence of cow dung and urea fertilizers on soil pH, OC, TN and microbial population under in a savanna Alfisols. The treatments consisted of three levels of cow dung (0, 3 and 6 tons/ha) and four rates of nitrogen (0, 50, 100 and 150 kg/ha) arranged in completely randomized design (CRD). Measurements of soil microbial population (SMP), soil pH, soil organic carbon, (OC), total nitrogen (TN) and yield parameters were conducted. The result revealed significant effect of CD on all the parameters except yield parameter. It also showed similar effect of N rates. The results indicates that a combination 3 tons/ha CD and 100 kg/ha of nitrogen fertilizer could sustain optimum sustainable for better soil condition and maize yield in the study area. It recommended that farmers should adopt the application of 3 tonnes/ha which is the carrying capacity of the soil alongside 100kg N/ha.

Introduction

Maize is a diverse crop that can grow in many agro-ecological zones of Nigeria. In recent times, maize has become an important crop in the savanna ecologies by taking over the position of traditional crops, such as millet and sorghum, by increase in productivity, with an average yield of between 3000 - 6000 kg per ha every year since 1960 (Leveau *et al.*, 2011). This trend of production maintains significant increase in recent years (FAO, 2018). For example, in 2014, Nigeria's average maize yield was 1.8 metric tons /ha (FAOSTAT, 2014 not on reference list) and ranked as one of the lowest among the top 10 maize producers in Africa, lagging behind countries such as Egypt and South Africa, where the yields are 7.7 metric tonnes/ha and 5.3 million tonnes/ha respectively (FAOSTAT, 2014). In 2018, the trend improved, when Nigeria produced about 10.2 metric tons/ha of maize, making it the highest producer in Africa (FAO, 2018). The current high production indicates that Nigeria can become the largest maize producer in Africa and one of the largest producers in the world without increasing the area currently used for its cultivation (FAO, 2018). Hence need to improve soil conditions for better yield and nutriment management for higher yield.

Soil microbial population is the living component of soil organic matter. It is considered as a sensitive indicator of soil quality and health (Burns *et al.*, 2013). Soil organic matter

serves as reservoir of soil N, P, K and S while microbial population takes part in nutrient transformations. Upon addition of inorganic and organic fertilizers, the microbial organism acts on and releases the nutrients into readily available form for plant uptake. They serve as major determinant of nutrient availability and may practically act as slow release fertilizer. The most important group of microorganism is bacteria and fungi. The combined group alone account for more than 90% of the carbon dioxide evolved during decomposition of forest litter (Schaefer 1991). They also play significant role in nutrient cycle and formation of soil aggregates. Use of organic fertilizers in the soil has great positive effect on soil bacterial community (Tayyab *et al.*, 2018). Cow dung contribute to harboring a rich microbial diversity which contains different species of bacteria, protozoa and yeast (Sawant *et al.*, 2007) and increase nitrogen, carbon, phosphorus and improve pH which in turn increases microbial biomass (Das *et al.*, 2017). Objective of the study is not stated. This is very important in order for the author to know the direction of the work. Basic components (pH, OC and TN) of the topic not adequately defined. Relevance to the study not stated.

MATERIALS AND METHODS

Screen House activities

The screen house experiment was conducted at the Screen house of the Department of Soil Science, Ahmadu Bello University, Zaria in the 2021 dry season. Prior to establishment of the experiment, a total of 144 kg of soil was collected from a virgin land that has been left fallow for over 10 year in the IAR Experimental Farm, ABU Zaria. The sampled was collected at the depth of 0-15cm and then taken to the laboratory, air dried, crushed and sieved through the 2 mm mesh sieve. The pot with capacity of 4 liters was filled with 4kg of the prepared soil and sub sample was taken for soil analysis. Cow dung was applied to the pots a week before planting while the nitrogen fertilizer was applied together with P and K a day to a planting. The mixed soil and cow dung were wetted to field capacity and left for a week before planting. Plastic receivers were placed under the pots to collect any seepage, which was returned back into the pots regularly. The urea, single super phosphate (SSP) and murate of photash (MOP) fertilizers were applied to each pot, watered and left over night before planting. A total of 6 maize seeds (Oba super 2 variety) were sown in each pot. After germination, the plants were thinned to 4 stands. The soil was maintained at field capacity through regular watering, which at least 2-3 days interval

Organic manure Collection and Preparation

The organic manure used for the screen house and field experiments were cow dung. The cow dung was obtained from the dairy cattle pen of the National Animal Production Research Institute (NAPRI), Shika, ABU Zaria.

Treatment and experimental design Treatment combinations as well as replications not stated in line with objectives

The treatments consisted of three levels of cow dung and four rates of nitrogen. The cow dung levels were 0, 3 and 6 tons/ha which were equivalent to 0, 6 and 12 g/pot for the 4 kg soil/pot, while the nitrogen rates was 0, 50, 100 and 150 kg/ha which were equivalent to 0, 0.22, 0.44 and 0.66 g/pot respectively. Completely randomized design (CRD) was adopted for the study.

Harvesting

The plants were harvested after 6 weeks of growth in the screen house. Each pot was harvested by cutting the plants at 1cm above the soil level. Drying and weighing of

harvested shoot was done. The harvested plant shoots were washed thoroughly with water and dried for 48 hours in an oven set at 65°C and weighed to obtain the dry matter yield (DMY).

Post harvest soil sampling and analysis

The soil from the pots was sampled for chemical analysis and estimation of microbial population after harvesting. The soil samples for chemical analysis were air dried, crushed and sieved through 2 mm, while the soil for microbial population analysis was taken to the laboratory and stored in the refrigerator prior to the analysis. The various parameters estimated after planting were soil pH, soil organic matter and soil total nitrogen. The chemical properties of the soil samples were determined by using the following methods: soil pH was determined electrometrically in a soil solution ratio of 1:2.5, using the glass electrode pH meter. The organic carbon was analysed by wet oxidation method of Walkley and Black as modified by (Nelson and Sommers, 1982) and Total nitrogen by the micro-Kjeldahl distillation procedure according to (Bremner, 1996 not listed among references).

Determination of soil microbial population

The microbial counts for bacteria and fungi were conducted on the samples collected by using the spread plate method in serial dilution as described by Sanders (2012). The readings for colony count of bacteria and fungi were taken. The method involved the following step by step procedure.

Serial dilution for sample

Series of 6 test tubes were prepared with each containing 9 ml of sterilized distilled water. 1 g of soil sample was added to the first test tube and shaken thoroughly with the shaking machine. After shaking, 1 ml of sample was pipetted and transferred to the next test tube and then shaken again. The procedure continued up to the 6th test tube.

Plating for bacteria media

Exactly 10 ml of the 6th test tube solution was pipetted and dropped at the center of the petri dish containing nutrient agar solution. L-shape glass spreader used to spread content of the petri dish, was sterilized before placing in the petri dish by immersion into ethanol and then flamed over a Bunsen burner. The sample was spread over the surface of the agar using the sterile L-shape glass spreader and the petri dish was then rotated underneath at angle of 45° at the same time. The plate containing the sample was incubated at 37°C for 24 hours. The colonies were counted after 24 hours of incubation (Sanders 2012).

Plating for fungi media

Exact 1.0 ml of the 3rd test tube solution was pipetted and drop at the center of the petri dish containing Sabouraud Dextrosen Agar (SDA) solution. L-shape glass spreader was immersed into the alcohol and then flamed over a Bunsen burner as done before. The sample was spread over the surface of the agar using the sterile glass spreader and the petri dish was rotate underneath at angle of 45° at the same time. The plate containing the sample was incubated for 37°C for 72 hours. The colony was counted after 72 hours of incubation (Sanders 2012).

Colony count calculation

$$\text{CFU/ml} = \frac{(\text{No of colonies} \times \text{Dilution factor})}{\text{Volume of culture plate}}$$

Where

CFU = colonies forming unit

No of colonies = number of bacteria / fungi colonies observed in the plate after incubation

Volume of culture plate = the volume of the sample aliquot applied to the plate (1.0ml)

Dilution factor = the test tubes sampled were 10^{-6} for bacteria and 10^{-3} for fungi.

Statistical Analysis of the Data

The data collected were subjected to Analysis of Variance (ANOVA) using SAS package (SAS, 1999). Significant means were separated using the Duncan Multiple Range (DMRT) at 5% level of probability.

RESULT AND DISCUSSION

There is no table showing the initial properties of the soil before the screenhouse experiment

Effect of Cow Dung and Nitrogen Fertilizer on Soil chemical properties

Soil pH

Author should state the basis for comparison since there is no objective of the study

Effect of cow dung on soil pH is presented in Table 1 this table does not present data on pH alone. All other parameters should be stated in this opening statement. The pH (H₂O) was neutral with CD application and increased significantly at 3 tons/ha of CD applied. So it was also with pH in (CaCl₂) though, mean differences with CD levels were not significant. This observed slight increase in pH with the application of treatments relative to the pre experimental pH conditions as earlier stated in line with the report of Lekasi *et al.* (2003) you need to state what the report is.

The effect of nitrogen rates on the cropped soils followed similar trend as with cow dung application. Significant effects were only up to 50 kg N/ha for pH (H₂O). The neutral pH conditions must have been due to the amendment used. This result is in agreement with findings of Tanimu *et al.* (2007) who observed an improvement in soil pH due to high metabolic activities this is not correct.

Table 1: Effect of cow dung and nitrogen rates on soil pH, organic carbon and total nitrogen of the screen house cropped soil

Treatments	pH H ₂ O	pH CaCl ₂	Organic Carbon (g kg ⁻¹)	Total Nitrogen (g/kg)
Cow dung (t/ha)				
0	7.19 ^b	6.89 ^b	0.45 ^c	0.18 ^a
3	7.39 ^a	6.93 ^a	0.51 ^b	0.16 ^b
6	7.34 ^a	6.93 ^a	0.64 ^a	0.17 ^a
Mean	7.31	6.92	0.53	0.17
S.E±	0.02**	0.01**	0.01**	0.03*
N rate (kg/ha)				
0	7.23 ^c	6.92 ^{ab}	0.67 ^a	0.14 ^c
50	7.44 ^a	6.92 ^{ab}	0.36 ^d	0.15 ^b
100	7.25 ^{bc}	6.89 ^b	0.63 ^b	0.19 ^a
150	7.31 ^b	6.94 ^a	0.47 ^c	0.19 ^a
Mean	7.31	6.92	0.53	0.17
S.E±	0.02**	0.01	0.01**	0.03**

Means follow by the same letter(s) in the same column are not significantly different at 5% level of probability, ***= highly significance. Table did not show the interaction effects of the two factors on the studied soil properties

Soil Organic Carbon (OC)

Effect of cow dung (CD) on soil organic carbon content revealed significant ($p \leq 0.05$) differences among treatments means (Table 1). All the treatments showed an increase in organic carbon content at harvest at 6 WAS compared to the initial organic carbon content. There was significant effect of treatments up to 6 tons/ha applied cow dung sentence should be rephrased. This is expected as increasing organic carbon content observed was in response to increasing rate of cow dung applied. This result is in line with finding of N'dayegamiye *et al.* (1989 reference too old), who noticed an increase in the soil organic carbon with increase in cow dung application and attributed this to factors such as decomposition, mineralization and metabolic activities of the compost at high level of CD application.

Effect of nitrogen rates on organic carbon content was significant, though erratic. Organic carbon generally declined with N application though not in a consistent manner. The highest organic carbon observed was in the control plot (0.67 g/kg) and declined with increase in N rates. The high OM accumulation in no nitrogen plots may be attributed to reduced yield and therefore low plant activity in the plots. This agrees with finding of Yan *et al.* (2007) who investigated high organic carbon content in control than in high nutrient amended treatments plots.

Soil total nitrogen (TN)

The effects of cow dung and nitrogen rate on total nitrogen content were significant ($p \leq 0.05$) (Table 1). Total nitrogen content was found to decrease consistently with increase in cow dung application. The most possible reason to this may be due to higher crop vigour with increase in organic matter application which stimulated higher nitrogen uptake with progressive depletion of soil N content. The result contradicts the findings of Leye, (2012) which stated that ?????.

The effect of N rates on soil N content was consistent and nitrogen content of the soils increased with N application. Effects were found to be significant only up to 100 kg N/ha. Higher N application rate may be lost through leaching. This result is consistent with the report of Leye, (2012) author should state the report of the referenced author for proper reportage .

Effect of Cow Dung and Nitrogen Rate on Soil Microbial Population

Result of the effect of cow dung on soil microbial populations are presented on Table 2. The result revealed significant ($p \leq 0.05$) differences among the treatments for both bacteria and fungi. There was consistent and significant increase in bacterial population in response to treatment with about a tenfold increase at 3 tonnes/ha over the control. Bacterial count declined at 6 tonnes/ha cow dung application. Fungal population was highest at 0 tonnes/ha cow dung (CD control) and declined with increase in CD application please note, decline in fungal population was not linear as such this sentence should be rephrased. The results show that 3 t/ha CD and 50 kg N/ha gave the best treatment combination for bacteria, and

3 t/ha CD without N application for the fungal growth. The positive effects of 3 t/ha CD on growth parameters (this was not presented in the study therefore it should not form the basis of comparison) in the screen house seems to be empirically explained, and emphasized the huge role of soil microbes in influencing the fertility of the soils. Similar result was found by Yan *et al.* (2000), who reported high microbial population at soil carbon contents below certain level and further application reduced the microbial population was this findings on bacteria or fungi. This findings should be specific.

The effect of nitrogen rates on the microbial population in the screen house is also shown in Table 2. Results revealed significant increase of bacterial content with increase in N rates with 50 kg N/ha stimulating over 6 fold increases over the control plot this statement is contradicting results presented in Table 1. Bacterial count declined with additional increase in N application rate above 50 kg N/ha this is not what you presented in Table 2. On the other hand, the response of fungal count to N rates was highly significant and response was observed to be significant up to 100 kg N/ha. Fungal count declined at 150 kg N/ha. This result agrees with the finding of Keiblinger *et al.* (2010), in their study on the ability of fungi over bacteria to survive certain levels of N application due to acidity induced by N fertilizer application

Table 2: Effect of cow dung and nitrogen rate on soil microbial population

Treatments	Bacteria (cfu/g)	Fungi (cfu/g)	Means follow by the same letter(s) in the same column are not
<u>Cow dung (t/ha)</u>			
0	2080×10 ⁵ b	7698.3 ^a	
3	2080×10 ⁶ a	4887.5 ^b	
6	6910×10 ⁵ c	5956.9 ^{ab}	
Mean	184×10 ⁶	6180	
S.E±	1307.14**	679.42*	
<u>N rate (kg/ha)</u>			
0	556×10 ⁵ d	5750 ^b	
50	384×10 ⁶ a	5880 ^b	
100	121×10 ⁶ c	9229 ^a	
150	176×10 ⁶ b	3864 ^b	
Mean	184×10 ⁶	6180	
S.E±	4575**	792.67**	

significantly different at 5% level of probability, *=significance **= highly significance

Interaction between the two factors not presented in the table.

Effect of Cow Dung and Nitrogen Fertilizer on Dry Matter Yield (DMY) Is this part of the objectives???

The results of the effect of cow dung and nitrogen rate on maize dry matter yield are presented on Table 3 Table 3 should come immediately after the discussion. The DMY obtained ranged from 6.46 to 7.69 g per pot. The effect of both cow dung and nitrogen rate on dry matter yield were not significant though yield levels increased marginally with

increase in applied treatments. Interaction effects were also not significant not shown on the table. This is in line with the finding of Sangoi *et al.*, (2001), where he reported that maize crop utilized only between 20 and 25 kg of N from soil per ton of grain produced.

Conclusion and recommendation

From the result obtained in this study, the 3 tonnes/ha CD is the carrying capacity for both maize and soil microbial population of the study area. The treatment combination of 3 tonnes/ha cow dung and 50 kg N/ha produced the best effect for soil microbial population while 3 tonnes/ha CD and 100 kgN/ha for maize in the study area. It is recommended that application of 3 tonnes/ha and 100 kg N/ha is enough to improve soil properties, microbial population and maize yield in the study area.

Table 3: Effect of cow dung and nitrogen rate on maize dry matter yield (DMY)

Treatments	DMY (g)
<u>Cow dung (t/ha)</u>	
0	6.88
3	7.55
6	7.65
Mean	7.36
S.E±	0.49
<u>N rate (kg/ha)</u>	
0	6.46
50	7.54
100	7.76
150	7.69
Mean	7.36
S.E±	0.57

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